

Agentic FEA of the Zhangjinggao Construction Catwalk:

Coordinate gate, 142-portal reconciliation, a §7.76 PK2 main deck,
and a singular tangent on a disconnected centre-line

Run `catwalk-main-deck-gate-f23d`
Repository `Hiram-test/model`, branch `cursor/catwalk-main-deck-gate-f23d`
Pull request `#19`
2026-08-25

Abstract

This paper records a complete, auditable agentic finite-element modelling run for the Zhangjinggao construction catwalk. A 77 MB millimetre centre-line STEP is parsed, mapped into the project convention $x = \text{chainage} - K16 + 876.000$, classified against drawing topology, coarsened, and written as a CalculiX deck. Floor-rope and portal-rope anchors are separate `*NSET`/`*BOUNDARY` families. Twenty-one cross-passages and 142 portal frames (71 stations per deck; Chinese classifier U+6980, not U+69C7) reconcile against the drawing station lists with zero insertions. Frozen deck `82548e6a3bd2612b6b39a08c313402b32a1961af6eba018158267906276ab6da` is not rewritten; its `TYPE=STRESS` card remains `ELSET` plus uniaxial. The writer now emits CalculiX 2.21 §7.76 rows: element, integration point, and six global second Piola–Kirchhoff components $S = \sigma n \otimes n$. The new main deck `41fb32225489b0c6f993d3a077ce9293d472e4ede5ff644ca170bebbb` has 204 208 legal rows. CalculiX 2.21 reads it, assembles 879 076 equations, and stops on a singular factorisation. The coarsened centre-line has 22 096 connected components. Four solver files exist without a converged increment. Frequency targets remain isolated. This paper certifies a new hashed §7.76 deck and a documented singular tangent—not a solved spectrum.

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1 Introduction

Related work. Automated CAD-to-FEM pipelines exist for buildings and bridges, but they usually inherit an already-consistent coordinate system and a single support family. Construction catwalks of a multi-span suspension bridge break that assumption: two walkways, two rope families with different physical anchor stations, discrete portal frames, and discrete cross-passages, all written in highway chainage. A companion theory archive (`catwalk-theory/thirteen-mode-models`, v1.2) already freezes the drawing topology and the isolation protocol for fourteen low-order families. That archive is a theoretical model family, not a CalculiX deck.

Method. The run follows `catwalk-fem/SKILL.md` and `PLAN.md`. Geometry is STEP-only. Materials, sag, and loads come from drawings and the check report. The coordinate gate is a hard stop: identity unless saddle-height evidence requires a shift, and never $X - x_{\min}$. Topology is reconciled against frozen drawing lists: 21 passages and $71 \times 2 = 142$ portal frames. Anchors are two families, not one mixed set. The writer emits a complete keyword deck. After the hash is frozen, CalculiX may be run only on a copy.

Experiment. Release `catwalk-attachment23-v2.0-s10-20260716` file `cw_S10_0716t050342_a4_centerlin` (SHA-256 `d03d01e38b823df5af4c1ff9b0b175fdfb87b097b9cda9a03af5d14e9c763344`, 139 991 trimmed curves, `SI_UNIT(.MILLI., .METRE.)`) is converted to metres. After identity mapping, $X \in [0, 4270.609] \text{ m}$ and $Z_{\max} = 350.312 \text{ m}$. The written deck has 51 896 nodes and 30 317 elements (7 702 117 bytes). CalculiX CrunchiX 2.21, executed on a copy of hash `82548e6a`, dies while reading the first initial-stress row.

Outlook. A later run may emit a CalculiX-legal stress card without silently mutating this hash. North physical anchors lie outside this STEP and remain proxies. Fourteen-mode comparison stays forbidden until a solved spectrum exists.

2 Related work and isolation protocol

Related work. Agentic FEA here means a scripted, auditable chain with frozen constants, not a chat that invents sections. Dhondt’s CalculiX is used as the open solver because its keyword deck is inspectable and its version is pinned. Abaqus-like ELSET-wide uniaxial prestress is a common authoring convenience; it is not the CalculiX 2.21 contract.

Method. Isolation rules, taken from theory v1.2 §1 and the skill file:

1. STEP supplies coordinates only.
2. Properties come from DRW-A/B, CALC-INPUT, STD, or a named ASSUMP.

3. TARGET-FREQ lives in `catwalk-fem/isolated/` and is not imported by `write_inp` or the solver deck.
4. Formed sag is $h = 227.300$ m ($340.600 - 113.300$). The 255.56 m main-cable control sag is forbidden in the deck (verified absent).
5. The two decks keep independent degrees of freedom; no mirror constraint.
6. Floor-rope anchors and portal-rope anchors are never unioned.
7. After SHA-256 82548e6a is frozen, the `.inp` is not rewritten to “make the solver happy”.

Experiment. The isolated file lists fourteen frequencies starting at 0.0296 Hz. A full-text search of the frozen deck finds neither 0.0296 nor 255.56. `write_inp.py` does not import `isolated/TARGET-FREQ.json`. The STEP sibling `cw_S10_0716t050342_a4_eq.db` is listed as a forbidden source in the heading.

Outlook. Isolation is a scientific claim, not a courtesy. Any later comparison table must load TARGET-FREQ only after the deck, the solve, and the mode-shape classification are frozen.

3 Charter, evidence grades, and frozen constants

Related work. Evidence grades follow theory v1.2: DRW-A/B, CALC-INPUT, STD, TEST, ASSUMP. Portal and passage stations are DRW-B dimension chains. Saddle coordinates are CALC-INPUT (check-report tables 1-5 / 1-9).

Method. The charter (`artifacts/analysis_charter.json`) freezes: project `zjg-catwalk`, run `catwalk-main-deck-gate-f23d`, SI internal units, and the four load cases LC-DEAD-PRESTRESS, LC-PERSONNEL-UNIFORM, LC-WIND-Y, LC-FREQ. The coordinate origin is the nominal four-span north end, not a physical anchor:

$$x = \text{chainage} - K16 + 876.000, \quad y \text{ upstream} \rightarrow \text{downstream}, \quad z \text{ up.}$$

Physical anchors stay in four named stations:

$$\begin{aligned} x_{\text{floor},N} &= -23.895 \text{ m } (K16 + 852.105), & x_{\text{floor},S} &= 4210.368 \text{ m } (K21 + 086.368), \\ x_{\text{portal},N} &= -44.909 \text{ m } (K16 + 831.091), & x_{\text{portal},S} &= 4225.700 \text{ m } (K21 + 101.700). \end{aligned}$$

Nominal four-span breaks $\{0, 660, 2960, 3677, 4180\}$ and physical saddles $\{666.679, 2953.321\}$ are registered together and are never silently mixed.

Experiment. The heading of the frozen deck states the convention in plain text. Node $x_{\min} = 0$ equals the transformed geometry $x_{\min}$. South floor and south portal selected means differ by 11.1 m.

Outlook. If a later STEP covers the north physical anchors, those four stations must still remain four sets.

4 Ingestion and classification (N02–N07)

Related work. A centre-line STEP of 139 991 trimmed curves is not a member-role model. Role assignment has to be recovered from geometry against drawing lattices.

Method. `parse_step` stream-parses `CARTESIAN_POINT` / `TRIMMED_CURVE`, checks the STEP SHA-256, and scales millimetres to metres. `classify_segments` labels floor ropes by the drawing $y = \pm(21.45 \pm (0.85 + 0.26k))$ lattice, long high members as portal/handrail ropes, short transverse members as `portal_or_beam`, and long inter-deck members as `cross_passage`.

Experiment. Raw counts: `floor_rope` 43 200, `portal_or_beam` 4 719, `portal_rope` 356, `cross_passage` 0, `short_other` 91 618. No single segment has $\Delta y \geq 15$ m, because each 49.655 m passage is tessellated into ≈ 1.7 m pieces. At every drawing passage X , node Y already spans $[-24.3, 24.3]$ m. Detection therefore uses Y -span, not a single long beam. Unclassified `short_other` is dropped from the coarsened deck. Longitudinal chains keep drawing stations and a 12 m target spacing.

Outlook. Portal-rope classification remains incomplete after coarsening (227 elements). That bound is reported in §9 and is not hidden by the 142 frame count.

5 Coordinate and topology gates (N07/N09)

Related work. A previous pull request shipped the pipeline and left `artifacts/` empty. An empty directory cannot be a passed coordinate gate. Topology counts must be recovered from the mesh against drawing lists, with insertions booked rather than invented silently.

Method. `infer_x_transform` scores identity, minus- $x_{\min}$, and minus raw chainage. High- Z histogram modes sit near portal clusters (≈ 700 m and ≈ 3023 m), not on the saddles. Alignment is therefore confirmed by local Z_{p90} within 12 m of $x = 666.679$ and $x = 2953.321$. Passages: 21 drawing stations ($3 + 13 + 3 + 2$). Portal frames: 71 stations $\times$ 2 decks = 142. The Chinese unit is U+6980 (classifier for frames); U+69C7 is a different character and is not used.

Table 1: Coordinate and topology gates read from `coord_gate.json`.

Check	Result	Evidence
Units	PASS	mm STEP $\rightarrow$ m nodes; $x_{\max} = 4270.609 < 20\,000$
Transform	PASS	identity, shift = 0
Saddle Z	PASS	north $Z_{p90} = 330.78$ m, south 324.85 m vs 340.6 m
Two decks	PASS	Y medians $+20.6 / -20.6$ m vs ± 21.45
Formed sag	PASS	geometry 214.18 m vs 227.30 m, $ \Delta = 13.12$ m
No $x_{\min}$ shift in <code>.inp</code>	PASS	node $x_{\min} = 0$ equals geometry
Passages	PASS	21/21, insertions 0
Portal frames	PASS	142/142, insertions 0

Experiment. Twenty-one passages close by Y -span at every DRW-B station (Table 2). Portal frames close station-by-station (Table 5): 11 north-span, 41 main-span, 11 south 717 m, and 8 south 503 m stations, each hit on both decks. Missing stations: 0. Inserted drawing beams: 0.

Outlook. The 13.12 m sag residual is inside the 15 m gate and is not a formed-line fit. Histogram modes must not be used as saddle evidence in later runs.

6 Materials, disjoint anchors, and cable seed (N08–N10)

Related work. Theory v1.2 §4 forbids using a floor-rope anchor as a portal-rope anchor. The two families differ by 21 m at the north end and by 15.332 m at the south end.

Table 2: Twenty-one cross-passages (DRW-B dimension chains).

No.	Label	x (m)	Chainage
1	P01	159	K17+035.000
2	P02	330	K17+206.000
3	P03	501	K17+377.000
4	P04	838	K17+714.000
5	P05	1009	K17+885.000
6	P06	1180	K18+056.000
7	P07	1351	K18+227.000
8	P08	1504	K18+380.000
9	P09	1657	K18+533.000
10	H1	1810	K18+686.000
11	H2	1963	K18+839.000
12	H3	2116	K18+992.000
13	H4	2269	K19+145.000
14	H5	2440	K19+316.000
15	H6	2611	K19+487.000
16	H7	2782	K19+658.000
17	H8	3138.5	K20+014.500
18	H9	3318.5	K20+194.500
19	H10	3498.5	K20+374.500
20	H11	3843	K20+719.000
21	H12	4014	K20+890.000

Method. Rope $E = 1.20 \times 10^{11}$ Pa, floor/portal $A = 1400.42 \times 10^{-6}$ m², $\mu = 12.038$ kg/m (CALC-INPUT / DRW-A). Steel $E = 2.06 \times 10^{11}$ Pa, $\rho = 7850$ kg/m³. Horizontal seed

$$H = \frac{wL^2}{8h}, \quad h = 227.300 \text{ m}, \quad L = 2286.642 \text{ m}.$$

Per deck, $w_{\text{floor}} = 2.766$ kN/m and $w_{\text{portal}} = 0.709$ kN/m give $H_{\text{floor}} = 7.954$ MN and $H_{\text{portal}} = 2.039$ MN. Axial seeds are $\sigma_{\text{floor}} = 3.549611 \times 10^8$ Pa and $\sigma_{\text{portal}} = 2.426295 \times 10^8$ Pa. L0 wave frequency $f_1 = \sqrt{g/32h} = 0.036719$ Hz is a magnitude gate only.

Floor and portal anchors use different node sets and two *BOUNDARY cards. A third card holds saddles and nominal ends.

Table 3: Disjoint anchor families. North physical anchors lie outside the STEP.

Family	Station	Target x (m)	Selected x_{mean}	Mode
Floor N	K16+852.105	-23.895	0.000	STEP north-end proxy
Floor S	K21+086.368	4210.368	4209.985	matched, $\Delta = 0.38$ m
Portal N	K16+831.091	-44.909	46.358	STEP north-end proxy
Portal S	K21+101.700	4225.700	4221.093	matched, $\Delta = 4.61$ m

Experiment. Intersection of N_FLOOR_ANCHOR (312 nodes) and N_PORTAL_ANCHOR (16 nodes) is empty. Three *BOUNDARY cards are present. South families remain 11.1 m apart in x .

Outlook. The deck does not invent nodes at negative x . A later STEP that includes the north anchorage must still keep the four families separate.

7 Load cases (N11)

Related work. CalculiX does not inherit *DLOAD/*CLOAD across steps. Each step must restate gravity and the active package.

Method. Per-deck packages are split equally onto 16 explicit floor ropes so that $\sum_i w_{\text{rope}} L_i = w_{\text{deck}} L_{\text{deck}}$.

Table 4: Load-case resultants close against wL on the coarsened floor mesh.

Case	Deck w	Per-rope w	Resultant
Extra dead	877.16 N/m	54.82 N/m	9.019 MN
Personnel	8.400 kN/m	525 N/m	86.371 MN
Wind +Y	0.50 kN/m	31.25 N/m	5.141 MN

Experiment. Floor truss length in the coarsened mesh is 164516 m (32 lines including sag and residual over-classified floor segments; ≈ 5141 m/line versus ≈ 4300 m expected). LC-FREQ requests 20 eigenvalues and does not contain 0.0296 Hz.

Outlook. The 20 % floor-line excess is a classification bound, not a live-load invention. It remains visible after the 21/142 topology audit closes.

8 Solver deck and the CalculiX 2.21 initial-stress gate (N12–N14)

Related work. CalculiX CrunchiX 2.21 (§6.11.7 and §7.76) requires, for TYPE=STRESS without USER: element number, integration point, and six global second Piola–Kirchhoff components. T3D2 follows B31 and is expanded to C3D8I (§6.2.35), so eight integration points are written. ELSET plus one axial number is not that contract. The previous G11 pre-solve gate did not cover this card.

Method. Frozen hash 82548e6a keeps

```
*INITIAL CONDITIONS, TYPE=STRESS
E_FLOOR_ROPE, 3.549611e+08
E_PORTAL_ROPE, 2.426295e+08
```

and is not patched. `write_inp.py` now emits $S = \sigma n \otimes n$ on eight integration points per floor/portal truss. The mesh is reread from the frozen deck and written to a *new* path `zjg_catwalk_ccx221.inp`. Gate IC-G-ccx221-pk2 rereads the text independently.

Experiment. The frozen copy still fails as given (exit 201, E_FLOOR_ROPE, 3.549611E+08, no equations, < 1 s). The new deck SHA-256 41fb32225489b0c6f993d3a077ce9293d472e4ede5ff644ca170bebbbbb has 204208 eight-field rows, zero ELSET-uniaxial rows, and $\text{tr}(S) = \sigma_{\text{floor}}$ on the first row. Twenty-seven automated gates PASS. CalculiX 2.21 reads the new copy, assembles 879076 equations, and reports `matrix found to be singular (spooles.out)`, exit 255, wall-clock 10.35 s. The four-file set exists (`.frd` 80 B, `.dat` 42 B, header-only `.sta/.cvg`). A connectivity audit finds 22096 connected components (largest 129 nodes). Pinning every original node makes the matrix non-singular, so the singularity is a disconnected centre-line mechanism, not the PK2 card.

Outlook. The lexical gate now covers §7.76. The next modelling step is connectivity, not another rewrite of the eight-field card, and not a silent edit of 82548e6a.

9 Zhangjingao results

Related work. The scientific object is one published centre-line STEP of one construction catwalk.

Method. Results are read back from both hashed decks, the gate JSON, the 142-frame ledger, the independent reread, and the CalculiX 2.21 logs. No displacement from a completed increment is reported.

Experiment.

1. Coordinate convention holds without subtracting $x_{\min}$.
2. 21 passages and 142 portal frames, insertions 0 and 0.
3. Anchor families disjoint ($312 \cap 16 = \emptyset$).
4. Sag seed 227.300 m; 255.56 and 0.0296 absent from both decks.
5. Frozen hash unchanged. New hash 41fb3222 submitted.
6. Five unit-test modules PASS.
7. New TYPE=STRESS is accepted; 879 076 equations; singular tangent.

Outlook. The supported claim is a §7.76-legal new main deck plus an untouched illegal witness hash. It is not a static solution and not a fourteen-mode match.

10 Discussion and conclusions

Related work. PR #18 shipped an empty artifacts/. PR #19 first deposited 82548e6a. This run changes the writer, submits 41fb3222, and leaves traces under catwalk-fem/eval/.

Method. Visible limits are bounds. Hard gates (identity, 21/142, disjoint anchors, §7.76 lexicon, frozen hash, forbidden sources) close. Singularity is a bound, not a reason to withhold the new deck.

Experiment. Bounds: north anchors are STEP-end proxies; portal-rope classification is incomplete (227 elements); geometric sag 214.18 vs 227.30 m; floor-line length $\approx 20\%$ long; 22 096 connected components; no converged increment; no spectrum.

Outlook. From the published STEP, under $x = \text{chainage} - K16 + 876.000$, a readable CalculiX deck exists; floor and portal anchors are separate; 21 passages and 142 frames reconcile with zero insertions. Frozen 82548e6a remains ELSET+uniaxial. New 41fb3222 emits element + integration point + six global PK2. CalculiX 2.21 reads it and assembles 879 076 equations, then fails on a singular factorisation caused by 22 096 disconnected components. No solved spectrum is claimed.

Reproduction

```
python3 catwalk-fem/tests/test_coord_gate.py
python3 catwalk-fem/tests/test_write_inp.py
python3 catwalk-fem/tests/test_reconcile.py
python3 catwalk-fem/tests/test_audit_frozen_deck.py
python3 catwalk-fem/tests/test_new_main_deck.py
python3 catwalk-fem/pipeline/emit_new_main_deck.py
sha256sum catwalk-fem/artifacts/zjg_catwalk_coarsened.inp
sha256sum catwalk-fem/artifacts/zjg_catwalk_ccx221.inp
pdflatex -interaction=nonstopmode zjg_catwalk_agentic_fea.tex
```

Do not add the 77MB STEP to git. Do not feed `isolated/TARGET-FREQ.json` to the writer. Do not rewrite `zjg_catwalk_coarsened.inp`. This run does not push.

Data availability

Artefacts live under `catwalk-fem/artifacts/` on branch `cursor/catwalk-main-deck-gate-f23d`. Frozen hash `82548e6a...276ab6da`. New main hash `41fb3222...bbca924a`. Evaluation traces: `catwalk-fem/eval/GROK_SELF_EVAL.md`.

References

- [1] G. Dhondt, *CalculiX CrunchiX USER’S MANUAL version 2.21*, 2023, §6.2.35 and §7.76.
- [2] Zhangjinggao catwalk fourteen-mode independent theory archive, v1.2, `catwalk-theory/thirteen-mode-models`.
- [3] `catwalk-fem/SKILL.md` and `catwalk-fem/PLAN.md`.
- [4] Release `catwalk-attachment23-v2.0-s10-20260716`, `cw_S10_0716t050342_a4_centerline.step`, SHA-256 `d03d01e3...763344`.

A Station-by-station 142-frame ledger

Generated from `artifacts/portal_142_ledger.json`. Every station is present on both decks. Counting unit: U+6980, not U+69C7.

Table 5: Station-by-station reconciliation of 142 portal frames (71 stations $\times$ 2 decks). Unit: frames (CJK U+6980), not U+69C7.

No.	x (m)	Chainage	Span	Up	Dn	n_{up}	n_{dn}	OK
1	45	K16+921.000	N 660 m	Y	Y	5	5	Y
2	102	K16+978.000	N 660 m	Y	Y	8	8	Y
3	159	K17+035.000	N 660 m	Y	Y	8	8	Y
4	216	K17+092.000	N 660 m	Y	Y	31	35	Y
5	273	K17+149.000	N 660 m	Y	Y	8	8	Y
6	330	K17+206.000	N 660 m	Y	Y	8	8	Y
7	387	K17+263.000	N 660 m	Y	Y	32	35	Y
8	444	K17+320.000	N 660 m	Y	Y	9	9	Y
9	501	K17+377.000	N 660 m	Y	Y	8	8	Y
10	558	K17+434.000	N 660 m	Y	Y	34	38	Y

Table 5: 142 portal frames (continued)

No.	x (m)	Chainage	Span	Up	Dn	n_{up}	n_{dn}	OK
11	615	K17+491.000	N 660 m	Y	Y	9	9	Y
12	724	K17+600.000	Main 2300 m	Y	Y	8	8	Y
13	781	K17+657.000	Main 2300 m	Y	Y	8	8	Y
14	838	K17+714.000	Main 2300 m	Y	Y	8	8	Y
15	895	K17+771.000	Main 2300 m	Y	Y	24	29	Y
16	952	K17+828.000	Main 2300 m	Y	Y	8	8	Y
17	1009	K17+885.000	Main 2300 m	Y	Y	9	9	Y
18	1066	K17+942.000	Main 2300 m	Y	Y	25	30	Y
19	1123	K17+999.000	Main 2300 m	Y	Y	8	8	Y
20	1180	K18+056.000	Main 2300 m	Y	Y	8	8	Y
21	1237	K18+113.000	Main 2300 m	Y	Y	25	26	Y
22	1294	K18+170.000	Main 2300 m	Y	Y	8	8	Y
23	1351	K18+227.000	Main 2300 m	Y	Y	8	8	Y
24	1402	K18+278.000	Main 2300 m	Y	Y	45	50	Y
25	1453	K18+329.000	Main 2300 m	Y	Y	10	10	Y
26	1504	K18+380.000	Main 2300 m	Y	Y	10	10	Y
27	1555	K18+431.000	Main 2300 m	Y	Y	45	50	Y
28	1606	K18+482.000	Main 2300 m	Y	Y	10	10	Y
29	1657	K18+533.000	Main 2300 m	Y	Y	10	10	Y
30	1708	K18+584.000	Main 2300 m	Y	Y	44	49	Y
31	1759	K18+635.000	Main 2300 m	Y	Y	10	10	Y
32	1810	K18+686.000	Main 2300 m	Y	Y	10	10	Y
33	1861	K18+737.000	Main 2300 m	Y	Y	45	50	Y
34	1912	K18+788.000	Main 2300 m	Y	Y	10	10	Y
35	1963	K18+839.000	Main 2300 m	Y	Y	10	10	Y
36	2014	K18+890.000	Main 2300 m	Y	Y	45	50	Y
37	2065	K18+941.000	Main 2300 m	Y	Y	10	10	Y
38	2116	K18+992.000	Main 2300 m	Y	Y	10	10	Y
39	2167	K19+043.000	Main 2300 m	Y	Y	45	50	Y
40	2218	K19+094.000	Main 2300 m	Y	Y	10	10	Y
41	2269	K19+145.000	Main 2300 m	Y	Y	10	10	Y
42	2326	K19+202.000	Main 2300 m	Y	Y	29	32	Y
43	2383	K19+259.000	Main 2300 m	Y	Y	8	8	Y
44	2440	K19+316.000	Main 2300 m	Y	Y	8	8	Y
45	2497	K19+373.000	Main 2300 m	Y	Y	30	33	Y
46	2554	K19+430.000	Main 2300 m	Y	Y	9	9	Y
47	2611	K19+487.000	Main 2300 m	Y	Y	8	8	Y
48	2668	K19+544.000	Main 2300 m	Y	Y	32	37	Y
49	2725	K19+601.000	Main 2300 m	Y	Y	8	8	Y
50	2782	K19+658.000	Main 2300 m	Y	Y	9	9	Y
51	2839	K19+715.000	Main 2300 m	Y	Y	32	37	Y
52	2896	K19+772.000	Main 2300 m	Y	Y	8	8	Y
53	3018.5	K19+894.500	S 717 m	Y	Y	8	8	Y
54	3078.5	K19+954.500	S 717 m	Y	Y	9	9	Y
55	3138.5	K20+014.500	S 717 m	Y	Y	8	8	Y
56	3198.5	K20+074.500	S 717 m	Y	Y	8	8	Y
57	3258.5	K20+134.500	S 717 m	Y	Y	8	8	Y
58	3318.5	K20+194.500	S 717 m	Y	Y	8	8	Y

Table 5: 142 portal frames (continued)

No.	x (m)	Chainage	Span	Up	Dn	n_{up}	n_{dn}	OK
59	3378.5	K20+254.500	S 717 m	Y	Y	9	9	Y
60	3438.5	K20+314.500	S 717 m	Y	Y	8	8	Y
61	3498.5	K20+374.500	S 717 m	Y	Y	9	9	Y
62	3558.5	K20+434.500	S 717 m	Y	Y	8	8	Y
63	3618.5	K20+494.500	S 717 m	Y	Y	8	8	Y
64	3729	K20+605.000	S 503 m	Y	Y	8	8	Y
65	3786	K20+662.000	S 503 m	Y	Y	8	8	Y
66	3843	K20+719.000	S 503 m	Y	Y	8	8	Y
67	3900	K20+776.000	S 503 m	Y	Y	23	24	Y
68	3957	K20+833.000	S 503 m	Y	Y	8	8	Y
69	4014	K20+890.000	S 503 m	Y	Y	8	8	Y
70	4071	K20+947.000	S 503 m	Y	Y	22	24	Y
71	4128	K21+004.000	S 503 m	Y	Y	8	8	Y

B CalculiX 2.21 transcripts

Frozen hash (unchanged):

```
*ERROR reading *INITIAL CONDITIONS. Card image:
E_FLOOR_ROPE,3.549611E+08
```

New hash:

```
number of equations
879076
matrix found to be singular
```

First legal stress row:

```
*INITIAL CONDITIONS, TYPE=STRESS
1, 1, 1.439957e+08, 0.000000e+00, 2.109655e+08, 0.000000e+00, 1.742932e+08,
0.000000e+00
```